

1 Explicit Instructions (Most Reliable in Chat)

Principle

LLMs respond strongly to **clear length constraints** stated directly in the prompt.

This is the simplest and most effective method when you do **not control an API**.

Examples

Limit by Words

Explain blockchain in **under 100 words**.

Limit by Sentences

Answer in **exactly 3 sentences**.

Limit by Format

Give a **maximum of 5 bullet points**.

Why This Works

The model internally optimizes toward:

- Instruction compliance
- Pattern matching with training data
- Reduced probability space due to constraints

Clear constraints dramatically improve output control.

2 Structural Output Constraints (Even Stronger)

Principle

Instead of requesting “short answers,” define a **strict output structure**.

Example

Respond using this format only:

Summary: (max 30 words)

Key Point: (1 sentence)

Why This Works

Structure:

- Reduces token entropy
 - Forces the model to fit within defined slots
 - Improves predictability
-

3 API-Level Token Limits (Hard Control)

Principle

When using an API, you can enforce a **hard generation cap**.

This is the **only guaranteed limiter**.

Python Example

```
response = client.chat.completions.create(  
    model="gpt-4o-mini",  
    messages=[...],  
    max_tokens=100  
)
```

What This Does

- Physically prevents output beyond the limit
 - Cuts off generation when reached
-

4 Understanding Tokens vs Words

Approximate Conversion

Tokens	Words
50 tokens	~35 words
100 tokens	~75 words
500 tokens	~375 words

Rule of Thumb

1 token $\approx$ **0.75 English words**

5 Advanced Control Techniques (Pro Level)

A. Combine Hard + Soft Limits

Best practice:

- Use `max_tokens`
- Also include **explicit length instructions**

This ensures:

- Compliance
 - Safety cap
-

B. Use Stop Sequences

Define a stopping pattern.

Example

```
stop = ["END"]
```

The model stops when it generates the defined sequence.

C. Temperature Reduction

Lower temperature produces:

- Shorter outputs
- More deterministic responses

Example

temperature = 0.2

Common Mistakes

Saying “be brief”

Too vague → unreliable.

Asking for “short explanation”

No measurable constraint.

Ignoring context limits

Input and output **share the same token budget**.

Best Practice Formula

For consistent output control:

Clear length instruction + structured format + max_tokens parameter

This combination provides **near-deterministic output sizing**.